Example 5.10 The line r passes through the points $(-2,\ 2)$ and $(5,\ 8)$ and the line s passes through the points $(-8,\ 7)$ and $(-2,\ 0)$. Is the line r perpendicular to s?

Solution The slope of line r is $m_1 = \dfrac{8-2}{5+2} = \dfrac{6}{7}$

The slope of line s is $m_2 = \dfrac{0-7}{-2+8} = \dfrac{-7}{6}$

The product of slopes $= \dfrac{6}{7} \times \dfrac{-7}{6} = -1$

That is, $m_1 m_2 = -1$

Therefore, the line r is perpendicular to line s.

Example 5.11 The line p passes through the points $(3,-2)$, $(12,4)$ and the line q passes through the points $(6,-2)$ and $(12,2)$. Is p parallel to q ?

Solution The slope of line p is $m_1 = \dfrac{4+2}{12-3} = \dfrac{6}{9} = \dfrac{2}{3}$

The slope of line q is $m_2 = \dfrac{2+2}{12-6} = \dfrac{4}{6} = \dfrac{2}{3}$

Thus, slope of line p = slope of line q.

Therefore, line p is parallel to the line q.

Example 5.12 Show that the points $(-2,5)$, $(6,-1)$ and $(2,2)$ are collinear.

Solution The vertices are $A(-2,5)$, $B(6,-1)$ and $C(2,2)$.

Slope of $AB = \dfrac{-1-5}{6+2} = \dfrac{-6}{8} = \dfrac{-3}{4}$

Slope of $BC = \dfrac{2+1}{2-6} = \dfrac{3}{-4} = \dfrac{-3}{4}$

We get, Slope of AB = Slope of BC

Therefore, the points $A,\ B,\ C$ all lie in a same straight line.

Hence the points A, B and C are collinear.

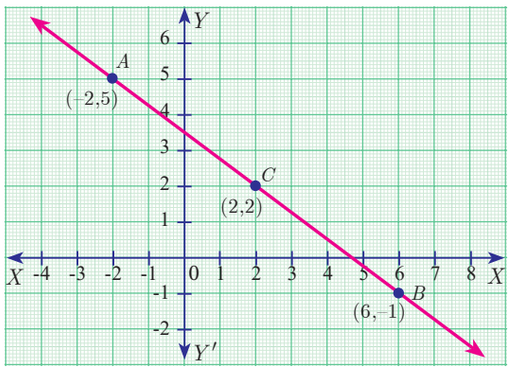


Fig. 5.23

Example 5.13 Let $A(1,-2)$, $B(6,-2)$, $C(5,1)$ and $D(2,1)$ be four points

(i) Find the slope of the line segments (a) AB (b) CD

(ii) Find the slope of the line segments (a) BC (b) AD

(iii) What can you deduce from your answer.

Solution (i) (a) Slope of $AB = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{-2+2}{6-1} = 0$

(b) Slope of $CD = \dfrac{1-1}{2-5} = \dfrac{0}{-3} = 0$

DO YOU KNOW?

If the slopes of both the pairs of opposite sides are equal then the quadrilateral is a parallelogram.

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